The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 November 2005.

INCH-POUND
MIL-PRF-19500/439F
18 August 2005
SUPERSEDING
MIL-PRF-19500/439E
23 February 2001

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER,
TYPES 2N5038 AND 2N5039, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1.1 Scope. This specification covers the performance requirements for NPN silicon, high-power transistors for use in high-speed power-switching applications. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurence are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (TO-3) and figure 2 (JANHC and JANKC).

1.3 Maximum ratings. T_A = +25°C (unless otherwise specified).

	P_T (1) at T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	W	V dc	V dc	V dc	V dc	A dc	°C
2N5038	140	150	90	7.0	5	20	-65 to +200
2N5039	140	125	75	7.0	5	20	-65 to +200

(1) Derate linearly 800 mW/°C for T_A > +25°C.

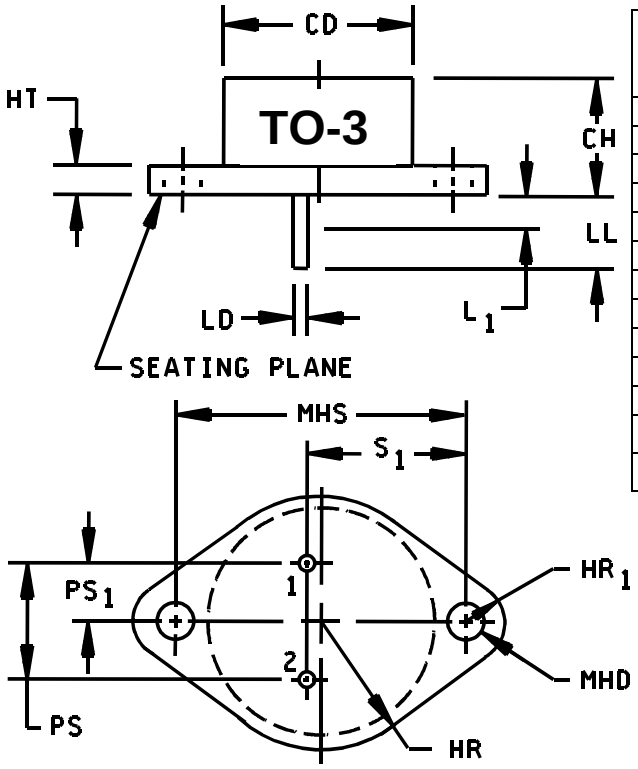
* 1.4 Primary electrical characteristics at T_A = +25°C.

Limit	h_{FE3} (1) V_{CE} = 5.0 V dc I_C = 12 A dc (2N5038) I_C = 10 A dc (2N5039)	$\lvert h_{fe} \rvert$ V_{CE} = 10 Vdc I_C = 2.0 A dc f = 5 MHz	$V_{CE(sat)1}$ (1) I_C = 12 A dc I_B = 1.2 A dc (2N5038) I_C = 10 A dc I_B = 1.0 A dc (2N5039)	C_{obo} V_{CB} = 10 Vdc I_E = 0 100 kHz ≤ f ≤ 1MHz	Pulse Response		$R_{\Theta JC}$ (2)
					t_{on}	t_{off}	
			V dc	pF	µs	µs	°C/W
Min	15	12					
Max		48	1.0	500	0.5	2.0	1.25

(1) Pulsed (see 4.5.1).
(2) See figure 3 for thermal impedance curve.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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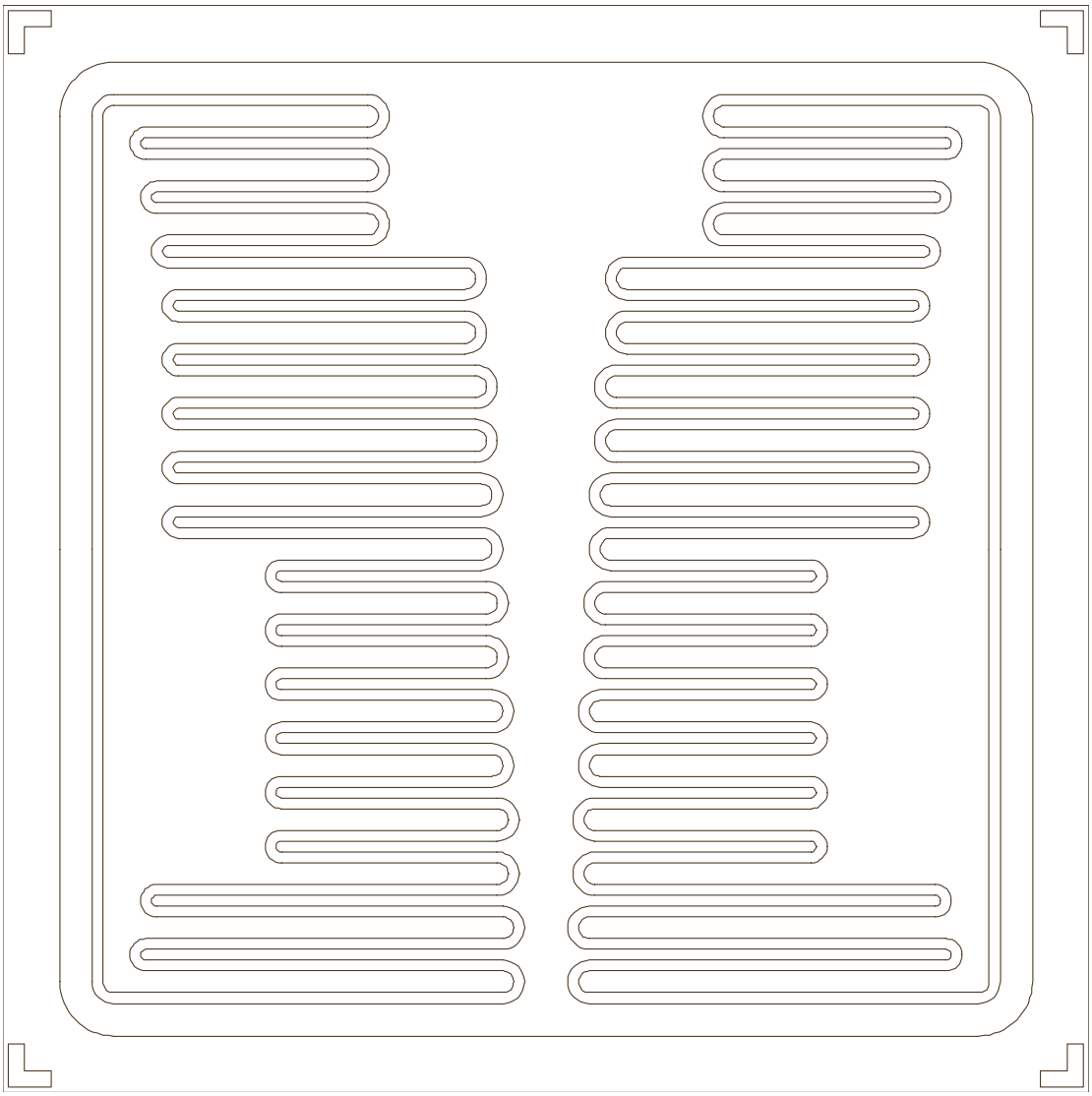


Symbol	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.22	
CH	.270	.380	6.86	9.65	
HR	.495	.525	12.57	13.33	4
HR_1	.131	.188	3.33	4.78	4
HT	.060	.135	1.52	3.43	
LD	.038	.053	0.97	1.35	4, 6
LL	.312	.500	7.92	12.70	
L_1		.050		1.27	6
MHD	.151	.165	3.84	4.19	4
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	3
PS_1	.205	.225	5.21	5.72	3
S_1	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. These dimensions should be measured at points .050 - .055 inch (1.27 mm - 1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
4. Two places.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Lead diameter shall not exceed twice LD within L_1.
7. Terminal 1 is emitter; terminal 2 is base; case is collector.

* FIGURE 1. Physical dimensions, (TO-3).

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NOTES:

1. Chip size: .200 X .200 inch (5.08 mm) ±.003 inch (0.08 mm).
2. Chip thickness: .015 inch (0.38 mm).
3. Top metal: Aluminum 30,000 Å min, 33,000 Å nominal.
4. Back metal: AlTiNiAu for solder mount 5,000A/2,000A/7,000A/5,000 Ang
   or
   Al/Ti/Ni/Ag 15,000/5,000/10,000/10,000.
5. Backside: Collector.
6. Bonding pad: B = .027 inch (0.69 mm) X .053 inch (1.35mm) ±.002 inch (0.05 mm).
   E = .023 inch (0.58 mm) X .059 inch (1.50 mm) ±.002 inch (0.05 mm).

* FIGURE 2. JANHC and JANKC (A-version die dimensions).

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-3) and figure 2 (die).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.2 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.3 Screening (JANTXV and JANTX levels). Screening shall be in accordance with table IV of MIL-PRF19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTX and JANTXV levels
3c (1)	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	Optional	Optional
9	I_{CEX1}, h_{FE2}, and h_{FE3}	I_{CEX1}
11	I_{CEX1}, h_{FE2}, and h_{FE3}; ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE3} = ±20 percent of the initial value.	I_{CEX1}, h_{FE2}, and h_{FE3}; ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13 (2)	Subgroup 2 and 3 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE3} = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE3} = ±20 percent of initial value.
14	Required	Required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5°C ±12.5°C; V_{CB} = 60 V dc ±5 V dc; $T_A \leq$ +100°C.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500.

* 4.3.3 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750, as applicable, using the guidelines in that method for determining I_H, t_H, I_M, and t_{MD}. The thermal impedance limit shall comply with the thermal impedance graph in figure 3 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750, as applicable. See subgroup 4 of table II herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table Via (JANS) and table VIb (JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirement shall be in accordance with 4.5.3 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A.
B4	1037	2,000 cycles; V_{CB} = 10 V dc; ΔT_J between cycles ≥ = +100°C; $T_{on} = T_{off} \geq$ 1 minute.
B5	1027	$V_{CE} \geq$ 20 V dc; P_T = 80 W at T_C = +100°C or adjusted as required by the chosen T_C to give an average lot T_J = +225°C.
B6	3131	See 4.5.3.
B7	3053	Load condition C; (unclamped inductive load) (see figure 4); T_C = +25°C; duty cycle ≤ 10 percent; R_s = 0.1Ω; $t_r = t_p \leq$ 500ns, for qualification and large lot QCI, sample size = 22, c = 0; for small lot QCI, sample size = 6, c = 0. TEST 1: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 2Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 20 A; L = 70 μH, 0.1Ω. TEST 2: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 40Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 4.5 A; L = 500 μH, 0.1Ω.

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* 4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV). Electrical end-points shall be in accordance with table I, subgroup 2, and 4.5.3 herein.

Subgroup	Method	Condition
B3	2037	Test condition A
B3	1037	2,000 cycles; V_{CB} = 10 V dc; ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off} \geq 1$ minute
B5	3131	See 4.5.3.
B7	3053	Load condition C; (unclamped inductive load) (see figure 4); T_C = +25°C; duty cycle ≤ 10 percent; $R_S = 0.1\Omega$; $t_r = t_p \leq 500$ ns, for qualification and large lot QCI, sample size = 22, c = 0; for small lot QCI, sample size = 6, c = 0. TEST 1: t_p = 5 ms (vary to obtain I_C); $R_{BB1} = 2\Omega$; V_{BB1} = 10 V dc; $R_{BB2} = 20\Omega$; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 20 A; L = 70 μH, 0.1Ω. TEST 2: t_p = 5 ms (vary to obtain I_C); $R_{BB1} = 40\Omega$; V_{BB1} = 10 V dc; $R_{BB2} = 20\Omega$; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 4.5 A; L = 500 μH, 0.1Ω.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with 4.5.3 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	Test condition B.
C2	2036	Test condition A, weight = 10 lbs., t = 15 s.
C6	1037	6,000 cycles; V_{CB} = 10 V dc; ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off} \geq 1$ minute.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) and delta measurements shall be in accordance with table I, subgroup 2 herein and 4.5.3 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. I_M measurement — 2 A.

b. V_{CE} measurement voltage — 20 V.

c. I_H collector heating current — 2.6 A (minimum).

d. V_H collector emitter heating voltage — 20 V (minimum).

e. t_H heating time — Steady-state (see method 3131 of MIL-STD-750).

f. t_{MD} measurement delay time — 20 μs.

g. t_{SW} sample window time — 10 μs maximum.

The maximum limit for $R_{\Theta JC}$ shall be 1.25°C/W.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection (1) (2) (3) (4)	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1	Collector to emitter voltage (saturated) 2N5038 2N5039	3071	 I_C = 12 A dc, I_B = 1.2 A dc I_C = 10 A dc, I_B = 1.0 A dc	$\Delta V_{CE(sat)1}$	50 mV dc change from initial value.		
2	Collector to emitter cutoff current 2N5038 2N5039	3041	Bias condition A; V_{BE} = -1.5 V dc V_{CE} = 100 V dc V_{CE} = 85 V dc	ΔI_{CEX1}	100 percent of initial value or 1μA dc, whichever is greater.		
3	Forward-current transfer ratio 2N5038 2N5039	3076	V_{CE} = 5.0 V dc; I_C = 12 A dc I_C = 10 A dc	Δh_{FE3}	±25 percent change from initial value.		

(1) Devices which exceed the table I limits, herein, for this test shall not be acceptable.
(2) The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
  a. Subgroup 4, see step 1 above.
  b. Subgroup 5, see steps 1, 2, and 3 above.
(3) The electrical measurements for table VIb, (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
  Subgroups 3 and 6, see steps 1, 2 and 3 above.
(4) The electrical measurements for table VII of MIL-PRF-19500 are as follows:
  Subgroup 6, see steps 1, 2, and 3 (JANS) above and step 3 (JAN, JANTX, and JANTXV) above.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\Theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc, pulsed (see 4.5.1)	V(BR)CEO			
2N5038				90		V dc
2N5039				75		V dc
Collector-emitter cutoff current	3041	Bias condition D	I_{CEO}			
2N5038		V_{CE} = 70 V dc			1	µA dc
2N5039		V_{CE} = 55 V dc			1	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5.0 V dc	I_{EBO}		100	µA dc
Collector-emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N5038		V_{CE} = 100 V dc			5	µA dc
2N5039		V_{CE} = 85 V dc			5	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}			
2N5038		V_{CE} = 150 V dc			1	µA dc
2N5039		V_{CE} = 125 V dc			1	µA dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 25 mA dc	$V_{(BR)EBO}$	7.0		V dc
Base to emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 5 V dc; pulsed (see 4.5.1)	V_{BE}			
2N5038		I_C = 12 A dc			1.8	V dc
2N5039		I_C = 10 A dc			1.8	V dc
Base to emitter voltage (saturated)	3066	Test condition A, I_C = 20 A dc; I_B = 5 A dc	$V_{BE(sat)}$		3.3	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N5038		I_C = 12 A dc, I_B = 1.2 A dc			1.0	V dc
2N5039		I_C = 10 A dc, I_B = 1.0 A dc			1.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = 20 A dc, I_B = 5.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE1}			
2N5038				50		
2N5039				30		
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.0 A dc; pulsed (see 4.5.1)	h_{FE2}			
2N5038				50	200	
2N5039				30	150	
Forward current transfer ratio	3076	V_{CE} = 5 V dc; pulsed (see 4.5.1)	h_{FE3}			
2N5038		I_C = 12 A dc		15		
2N5039		I_C = 10 A dc		15		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX2}			
2N5038		V_{CE} = 100 V dc			100	µA dc
2N5039		V_{CE} = 85 V dc			100	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; pulsed (see 4.5.1),	h_{FE4}			
2N5038		I_C = 12 A dc		10		
2N5039		I_C = 10 A dc		10		
Subgroup 4						
Magnitude small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 2 A dc; f = 5 MHz	$\lvert h_{FE} \rvert$	12	48	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0 A dc; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		500	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Pulse response transfer ratio	3251	Test condition A except test circuit and pulse requirements in accordance with figure 5 herein.				
Turn on time		V_{CC} = 30 V dc	t_{on}			
2N5038		I_C = 12 A dc, I_{B1} = 1.2 A dc			0.5	µs
2N5039		I_C = 10 A dc, I_{B1} = 1.0 A dc			0.5	µs
Turn off time		V_{CC} = 30 V dc	t_{off}			
2N5038		I_C = 12 A dc, I_{B1} = $-I_{B2}$ = 1.2 A dc			2.0	µs
2N5039		I_C = 10 A dc, I_{B1} = $-I_{B2}$ = 1 A dc			2.0	µs
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C, t = 1 s, 1 cycle, see figure 6				
Test 1		I_C = 5 A dc V_{CE} = 28 V dc				
Test 2		I_C = 0.9 A dc V_{CE} = 45 V dc				
Test 3		I_C = 20 A dc V_{CE} = 7.0 V dc				
Test 4 (2N5038 only)		I_C = 0.23 A dc V_{CE} = 90 V dc				
Test 5 (2N5039 only)		I_C = 0.32 A dc V_{CE} = 75 V dc				
Safe operating area (clamped switching)	3053	T_C = +25°C, I_C = 20 A dc; (see figures 7 and 8)				
2N5038		Clamp voltage = 90 V dc				
2N5039		Clamp voltage = 75 V dc				
		(Device fails if clamp voltage is not reached)				
Electrical measurements		See 4.5.3, steps 1 and 3				
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

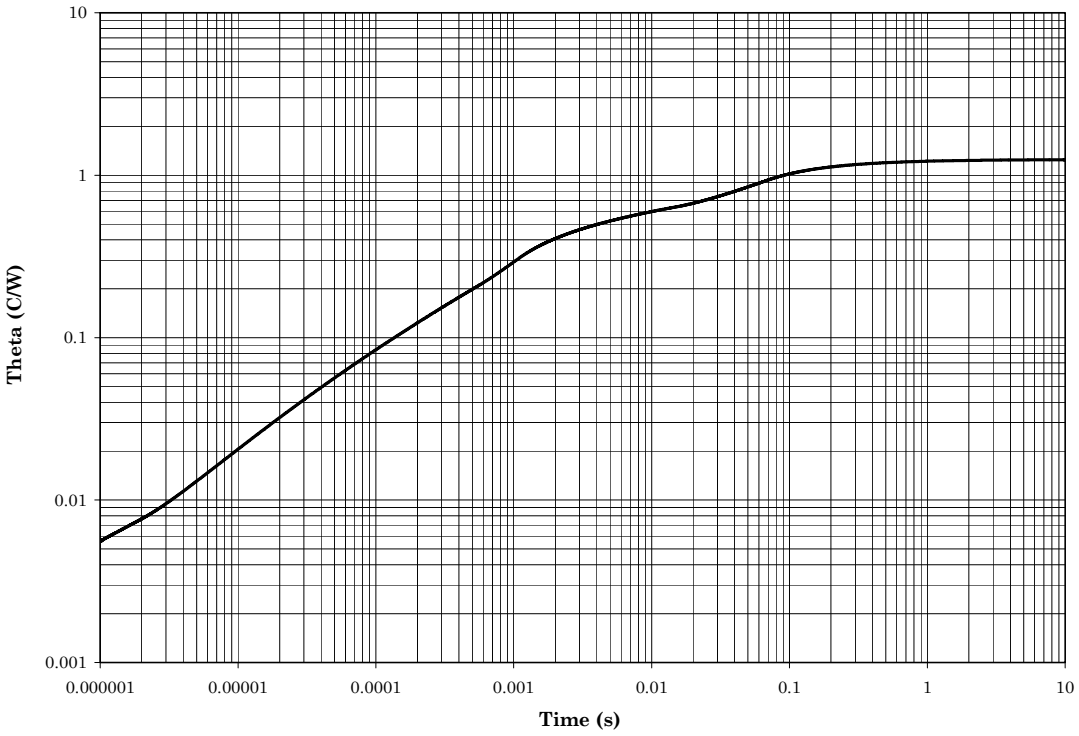
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* TABLE II. Group E inspection (all quality levels) for qualification and re-qualification only.

Inspection	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles	
Hermetic seal Fine leak Gross leak	1071	Test condition G or H, test condition C or D	
Electrical measurements		Table I, subgroup 2 and 4.5.3 herein. See steps 1, 2 and 3	
Subgroup 2			45 devices c = 0
High temperature reverse bias	1039	Condition A; 1,000 hours	
Electrical measurements		Table I, subgroup 2 and 4.5.3 herein. See step 2.	
Subgroup 3			3 devices c = 0
Not applicable			
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their qualification lot average and design-maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
Subgroups 5 and 6			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	

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2N5038, 2N5039 T_L = 25°C
Maximum Thermal Impedance Plots



$R_{\Theta JX}$ = 1.25°C/W
$Z_{\Theta JX}$ = 0.6°C/W at 10ms
$Z_{\Theta JX}$ = 0.75°C/W at 30ms

* FIGURE 3. Thermal impedance.

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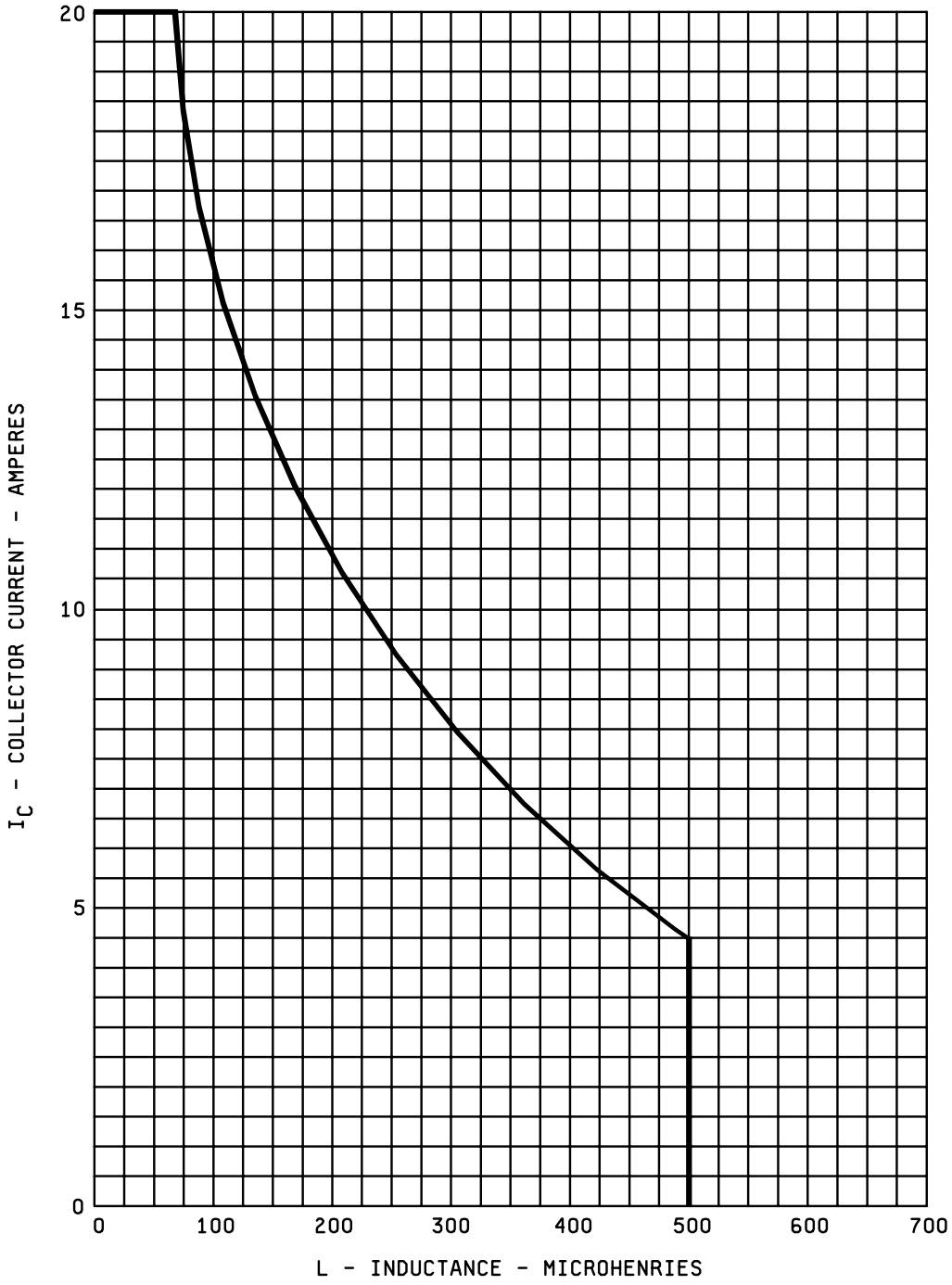
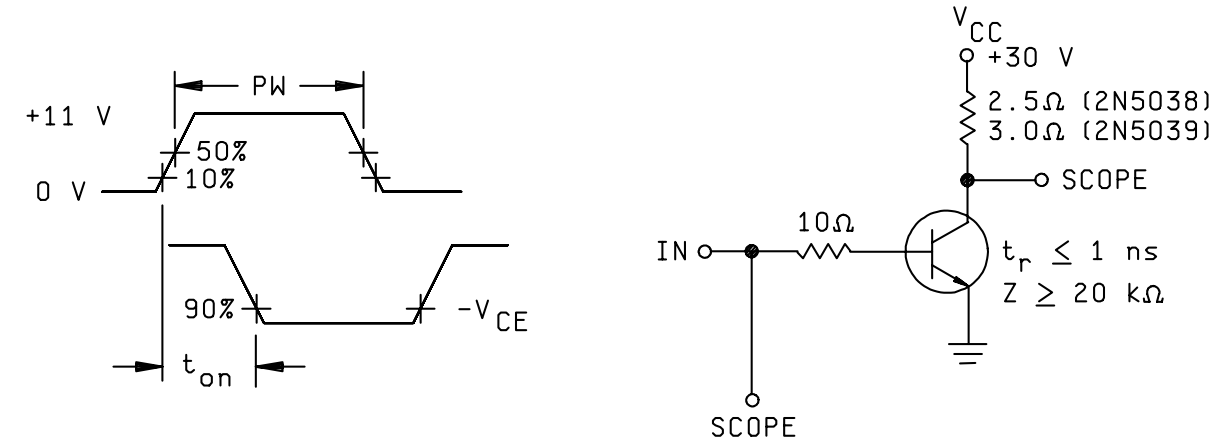
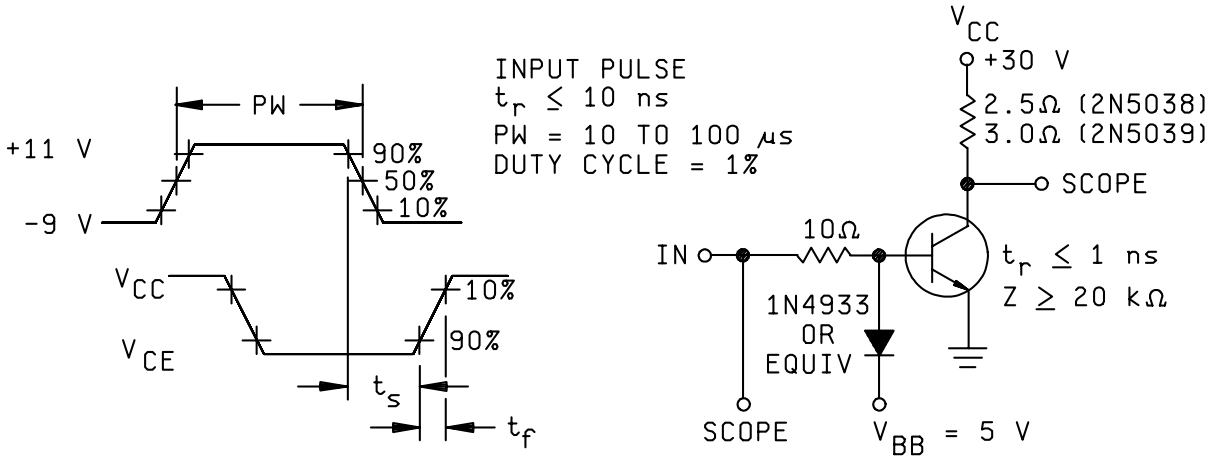


FIGURE 4. Safe operating area for switching between saturation and cutoff - unclamped inductive load.

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-ON (t_{off}) TIME TEST CIRCUIT

FIGURE 5. Switching time test circuits.

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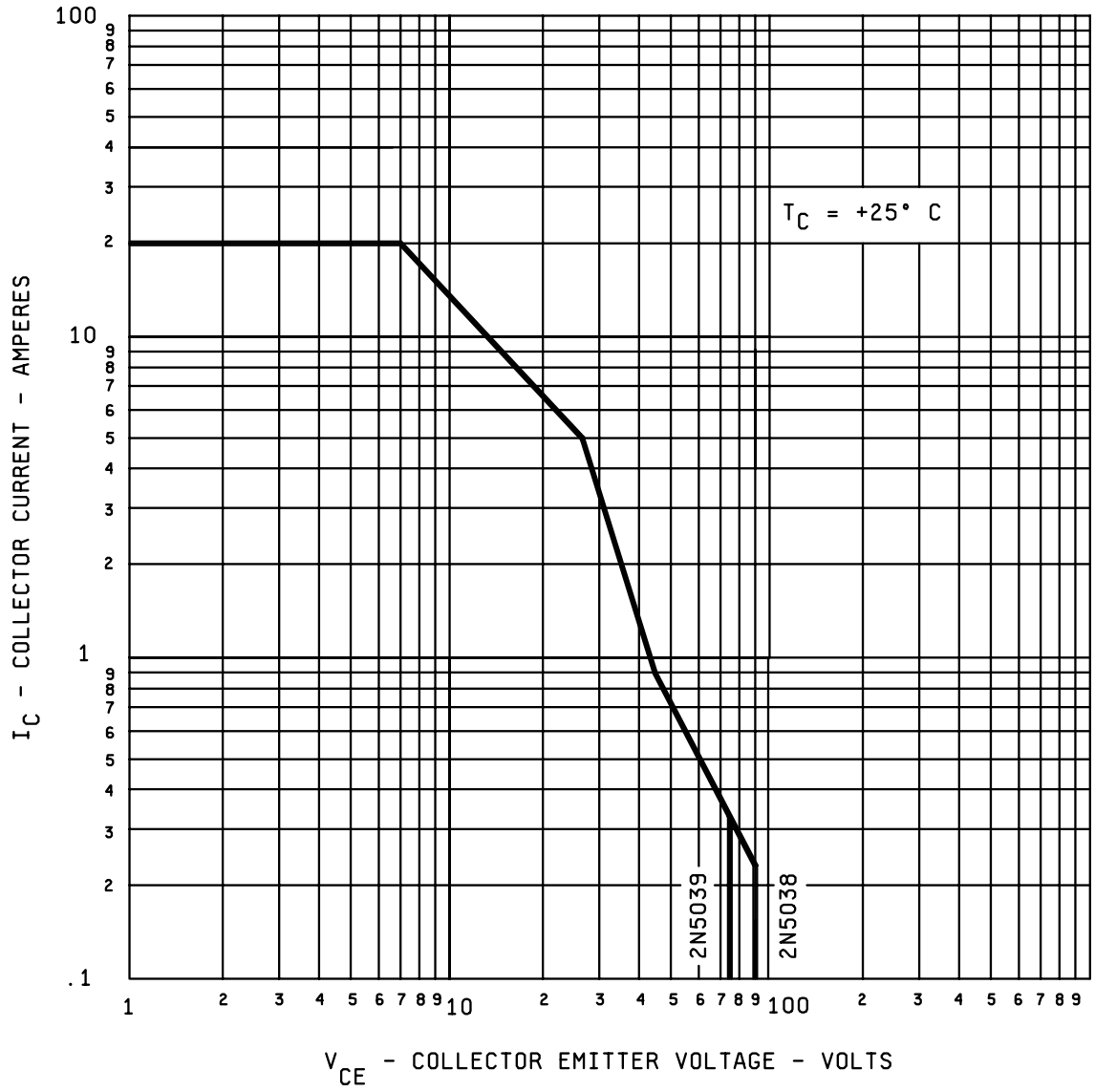


FIGURE 6. Maximum safe operating area graph (continuous dc).

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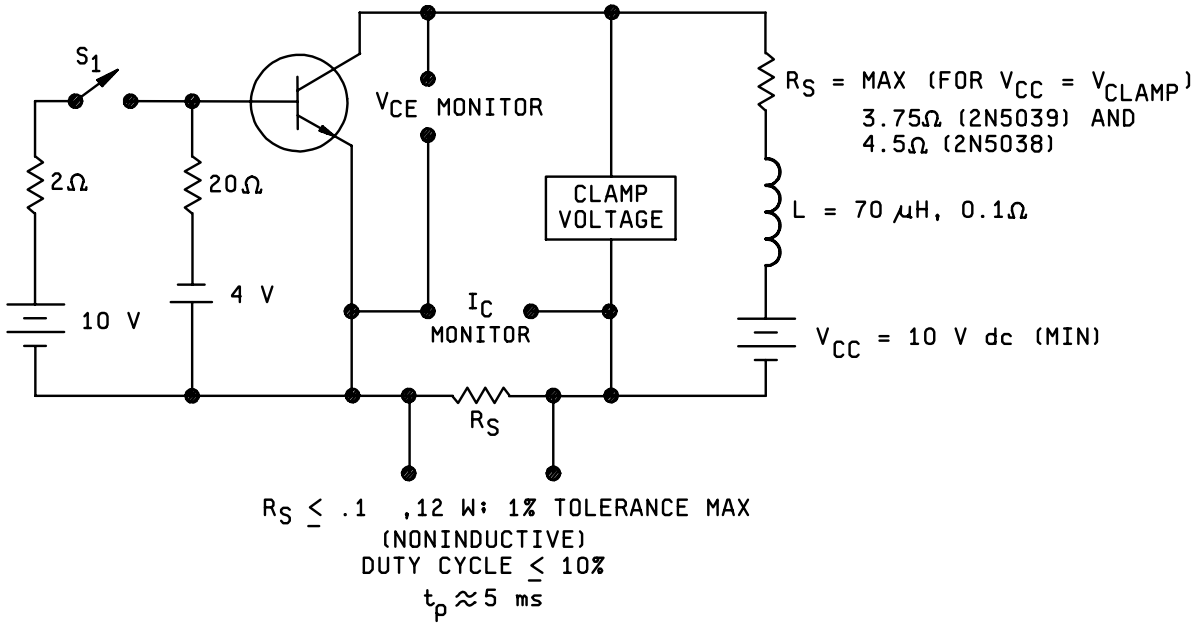


FIGURE 7. Clamped inductive sweep test circuit.

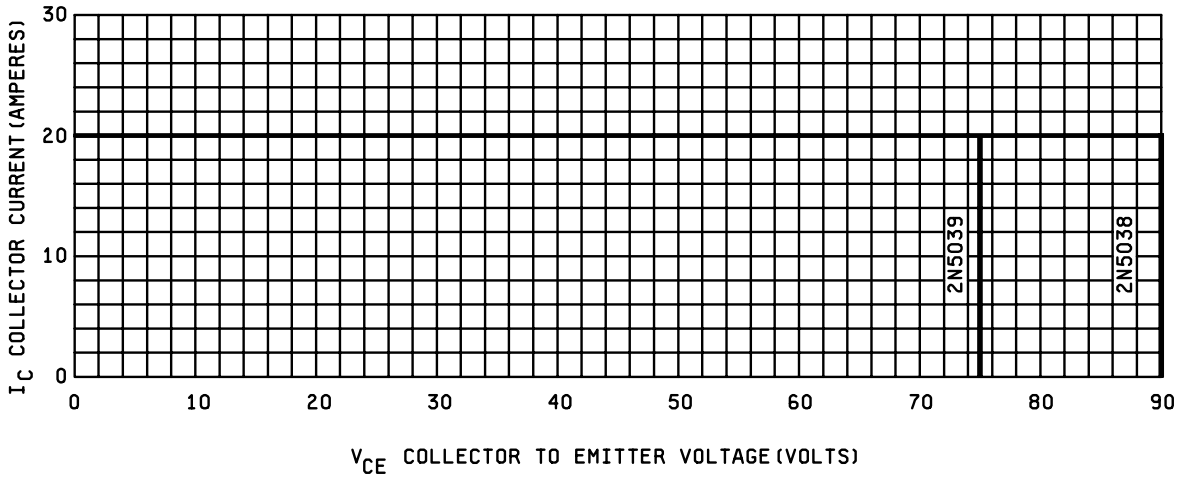


FIGURE 8. Safe operating area for switching between saturation and cutoff - clamped inductive load.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

* 6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5038) will be identified on the QPL.

Die ordering information	
PIN	Manufacturer
	43611
2N5038 2N5039	JANHCA2N5038 JANHCA2N5039
2N5038 2N5039	JANKCA2N5038 JANKCA2N5039

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2928)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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